Ex. 10 - Convoy

1) As for ex 3, a controller for our robot R_A would be:

$$u = A^{-1}(x).V \quad \text{with} \quad A(x) = \begin{pmatrix} -v_a \cos\theta_a & \cos\theta_a \\ v_a \sin\theta_a & \sin\theta_a \end{pmatrix}$$

and $V = (\hat{y} - y) + 2(\dot{\hat{y}} - \dot{y}) + \ddot{\hat{y}}$ where $y = \begin{pmatrix} x_a \\ y_a \end{pmatrix}$ and $\hat{y} = \begin{pmatrix} \hat{x}_a \\ \hat{y}_a \end{pmatrix}$

Knowing $\hat{y} = \begin{pmatrix} L_x \sin\omega t \\ L_y \cos\omega t \end{pmatrix}$, we have: $\dot{\hat{y}} = \begin{pmatrix} \omega L_x \cos\omega t \\ -\omega L_y \sin\omega t \end{pmatrix}$

$$\ddot{\hat{y}} = \begin{pmatrix} -\omega^2 L_x \sin\omega t \\ -\omega^2 L_y \cos(\omega t) \end{pmatrix}$$

2) If the distance between each robot should be d, that means the robot number i is at $d \times i$ from R_A. This is the curvilinear value of each robot R_i : $s_i = d \times i$

Every distance d_s, we keep the state X_A in a new column of S matrix.

We call k the number of d_s steps to reach R_i from R_A.

Thus, $s_i = k.d_s = d \times i$

So, if $S = [X_{A_1} X_{A_2} \cdots X_{A_j} \ldots X_{A_{m-1}} X_{A_m}]$ and j is the ~~current~~ last saved state of R_A,

so R_i should follow the state $\underline{j = m - k}$ with $k = \frac{d \times i}{d_s}$

Assuming we want the speed of each R_i the same as the current one of R_A, we will choose $v_i(t) = v_A(t)$

So we have $\hat{y}_i = \begin{pmatrix} x_{a_j} \\ y_{a_j} \end{pmatrix}$ and $\dot{\hat{y}}_i = \begin{pmatrix} v_{a_m} \cos\theta_{a_j} \\ v_{a_m} \sin\theta_{a_j} \end{pmatrix}$

As we don't have access to $\dot{\hat{y}}_i$ and $\ddot{\hat{y}}_i$, we could either approximate them with dt, or neglect them in the control law. So, the new control law ~~with~~ with poles equal to -1 would be:

$u = A^{-1}(x)V$ with $V = (\hat{y} - y) + (\dot{\hat{y}} - \dot{y})$